Project Documentation: ESP32 4WD Line Follower AGV

Overview: Engineering, mechanical modeling, hardware wiring, and software implementation of an Autonomous Guided Vehicle (AGV) line-following robot powered by an ESP32 microcontroller, 4WD skid-steering chassis, and TCRT5000 optical sensors.

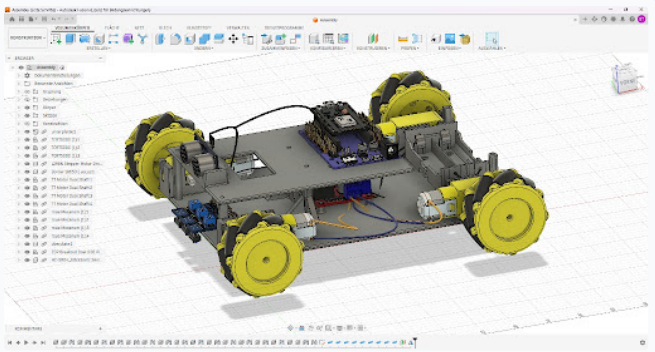


Fig 1: CAD 3D Assembly in Autodesk Fusion 360

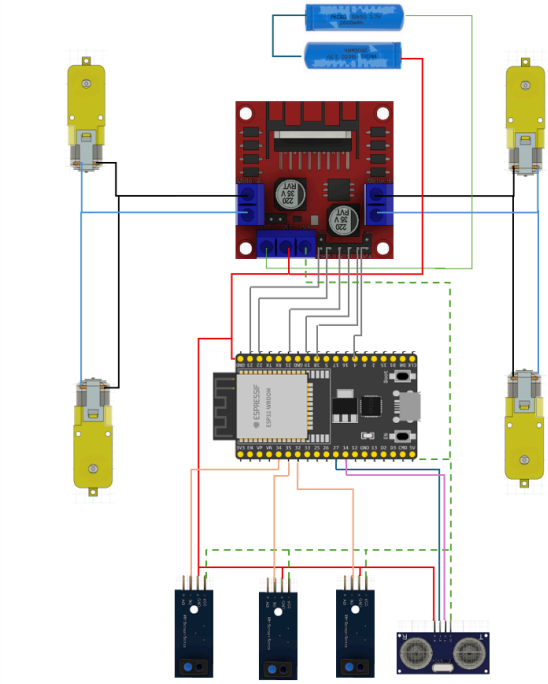


Fig 2: Complete Electrical Wiring Diagram

1. MECHANICAL DESIGN & CAD ASSEMBLY (AUTODESK FUSION 360)

The structural layout was completely modeled in **Autodesk Fusion 360** utilizing a modular multi-tier chassis architecture designed for optimal center-of-gravity and component accessibility:

- **Lower Plate Tier:** Holds four 1:48 TT gearhead DC motors paired in parallel, the L298N dual H-bridge motor driver module, and the front sensor array mounting bracket for precise ground-clearance positioning.
- **Upper Deck Tier:** Houses the main ESP32 breakout board, dual 18650 Li-ion battery holder, and an elevated mount for the HC-SR04 ultrasonic distance sensor.
- **Chassis & Drive Mechanics:** Configured for 4WD skid-steering. While providing high traction, four fixed wheels induce lateral tire friction during turns, requiring careful software compensation.

2. ELECTRONICS & CIRCUIT SCHEMATICS

The electrical system connects control logic, sensory input, and power distribution cleanly to ensure noise isolation and reliable current delivery:

- **Microcontroller Unit (MCU):** ESP32 WROOM-32 mounted on a 30-pin expansion breakout board to simplify pin accessibility.
- **Power Electronics:** L298N motor driver module delivering high-current DC power to the left and right motor banks.
- **Sensor Array:** 3x TCRT5000 infrared reflectance sensors mounted on a front array (20 mm center-to-center spacing) alongside an HC-SR04 ultrasonic sonar sensor for collision avoidance.
- **Power Supply:** 2x 18650 3.7V Li-ion batteries connected in series (~7.4V nominal output) providing raw voltage directly to the L298N power terminal, which regulates a 5V rail to power the ESP32 and logic sensors.

3. SOFTWARE ARCHITECTURE & CONTROL LOGIC

Written in C++ using the Arduino framework, the code implements soft-steering differential logic for smooth trajectory corrections:

- **Soft Steering Algorithm:** Instead of stopping inner wheels on course deviation, the code dynamically varies speed deltas between sides to maintain forward momentum.
- **Sensor Reading Logic:** The central TCRT5000 maintains straight trajectory when active over the line. Triggering the left or right outer sensor immediately initiates a proportional speed adjustment.

4. ENGINEERING CHALLENGES & SOLUTIONS

- **ESP32 GPIO Selection:** Initial crashes occurred due to binding motor PWM channels to strapping pins (GPIO 0, 2, 12, 15). Re-routing to dedicated output-safe pins solved boot stability issues.
- **Sensor Array Calibration & Track Tape Width:** Narrow 15mm electrical tape caused line drops during fast turns. Matching the track width to **25 mm black tape** matched the 20 mm sensor spacing perfectly.
- **Motor Stall Thresholds & Speed Tuning:** TT-motors stalled below PWM 120. Through extensive iterative testing, an optimal balance of **PWM 170 (straight)** and **PWM 230/120 (differential turning)** was established to overcome 4WD tire friction without slipping.

5. PROJECT CONCLUSION & OUTLOOK

Combining custom CAD design in Fusion 360, stable ESP32 pin allocation, and empirical PWM speed tuning resulted in a robust 4WD AGV platform. The complete codebase and documentation are now fully versioned and uploaded to GitHub.